

AMPLITUDE MODULES, FUSED SUPPORT, AND A COEFFICIENT-BAUTIN BOUND FOR POLYNOMIAL PHASES

ANONYMOUS

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ABSTRACT. We formulate two Atlas-derived amplitude-classification theorem candidates and one externally motivated ordinary coefficient-Bautin theorem candidate. We first reconstruct the load-bearing polynomial-amplitude identity from the *Cyclicity Support-Fusion Atlas* [14] self-containedly: a formal moment transform identifies the twisted quotient with distinct Kummer channels, a corrected falling-factorial Vandermonde argument computes cyclic rank, and Fourier inversion identifies the same channels with primitive root values. We then derive a general monodromy-block decomposition: central monodromy projectors descend to free polynomial modules, and every nonzero block component activates its entire Fourier-character block. For every nondegenerate Dickson phase $D_d(x, a)$, meaning $a \neq 0$, we then give a free $\mathbf{C}[D_d]$ -module decomposition of arbitrary polynomial primitives and prove the exact rank formula $q = 2N + \varepsilon$. This yields a parity classification: odd degree realizes precisely the even ranks, while even degree realizes every rank. Second, for the exceptional Ritt collision $P = (x^4 + x)^3$, we exhibit three explicit fused amplitude modules and prove that every nonzero element of each activates exactly three channels. Together with the two composition channels, these modules partition all eleven non-trace residues and realize every rank. Third, for every degree- d polynomial phase, every zero-cycle, and polynomial perturbations of arbitrary fixed degree, we prove that the ordinary coefficient Bautin ideal of the zero-cycle displacement is generated by Melnikov orders one through $d - 1$. The proof combines an all-order Lagrange–Bürmann identity, injectivity of differentiation on zero-cycle polynomial integrals, and a Cayley–Hamilton recurrence. Writing $b_{\text{coeff}}(m)$ for this explicit ordinary coefficient index, we obtain $b_{\text{coeff}}(m) \leq m - 1$. If the terse index $b(m)$ in [17] is intended in this coefficient sense, then the same bound answers its Problem 8.4. The $d - 1$ cutoff is sharp on the complexified cycle module. For integral cycles we prove only the lower witness d/p , where p is the least prime divisor of d . As supporting infrastructure, we also identify Atlas support with the minimal order of an operator annihilating the complete rapid-decay period system and retain corrected cycle-specific block bounds. Exact finite checks support the algebraic arguments but are not substitutes for proof.

Keywords: polynomial exponential periods; twisted de Rham modules; inverse-at-infinity support; Dickson polynomials; polynomial monodromy; creative telescoping; zero-cycles; Bautin ideals; Melnikov functions.

INTRODUCTION

Let $P \in \mathbf{C}[x]$ be a polynomial phase, let $A \in \mathbf{C}[x]$ be an amplitude, and consider rapid-decay exponential periods

$$I_{\Gamma, A}(s) = \int_{\Gamma} A(x) e^{sP(x)} dx.$$

The number and structure of independent channels activated by A govern both the geometry of the primitive G , where $G' = A$, and the differential equations satisfied by the complete period

system. The Cyclicity Support-Fusion Atlas [14] proposes a particularly sharp bridge: cyclic rank equals the number of non-trace inverse-series residues and also the reduced span of the primitive values on a generic fibre. Section 2 reconstructs that bridge as a theorem within this manuscript rather than importing it as a premise.

The Atlas leaves an open direction around classification by induced rank, support fusion and period order. Problems A and B below are two explicit amplitude-classification subproblems of that Atlas direction. Problem C is instead an ordinary coefficient-Bautin theorem candidate motivated by the externally posed Problem 8.4 of [17]. The period-order construction is supporting infrastructure, not a fourth headline claim.

Problem A. Classify the cyclic rank of every polynomial amplitude for the nondegenerate Dickson phase $D_d(x, a)$, with $a \neq 0$, uniformly in degree and parameter.

Problem B. Classify every polynomial amplitude for the degree-twelve exceptional Ritt collision $P = (x^4 + x)^3$, including hidden cancellation inside the fused blocks.

Problem C. For phases and perturbations of degree at most m , and every zero-cycle of the phase, bound the ordinary coefficient Bautin index of the displacement uniformly in the phase, perturbation and cycle.

Support-to-order bridge. Given arbitrary $P, A \in \mathbf{Q}[x]$, compute a primitive scalar operator for every rapid-decay period, prove its complete-system minimal order, predict that order from inverse support, and bound the order for a specified ray-cycle.

Our answers have a common pattern. We first identify the active inverse residues with the cyclic summands of the twisted de Rham quotient. We then use monodromy and explicit polynomial modules to classify those residues in the two structured families, use exact quotient reduction to construct the period operator, and use a second quotient algebra to control all Melnikov orders. The reconstructed support proof follows the Atlas proof architecture and was produced by the same research system, so it is self-contained but not an independent validation of the Atlas.

The organizing principle is finite-fibre algebra. On the exponential-period side, the rank- $(d - 1)$ twisted de Rham quotient turns the infinite sequence $[A dx], [PA dx], \dots$ into a finite cyclic-module problem. On the zero-cycle side, the rank- d algebra $R[t, x]/(f - t)$ turns the infinite power-moment sequence into a finite characteristic recurrence. Monodromy then supplies the missing information: Fourier support in the first quotient and injectivity of differentiation in the second. The two constructions are different instances of the same finite-cover mechanism rather than unrelated applications.

Main results. For an arbitrary degree- d phase, the inertia d -cycle has simple Fourier spectrum. The monodromy permutation module is therefore multiplicity-free and splits into character blocks. Using the reconstructed root-span/support equality, central projectors descend to free $\mathbf{C}[P]$ -modules, and the attainable supports are exactly the unions of these blocks. This supplies the organizing principle for the two explicit classifications below.

For $a \neq 0$, every primitive has a unique Dickson expansion in paired modules. Each nonzero pair activates both conjugate residue channels, because losing either would force a nonsplit quadratic element into the rational base field. This proves

$$q(D_d, G' dx) = 2N(G) + \varepsilon(G).$$

For $Q = x^4 + x$ and $P = Q^3$, we construct three rank-three modules

$$\begin{aligned} M_0 &= \langle x, x^7 + \frac{7}{4}x^4, x^{10} - \frac{5}{2}x^4 \rangle, \\ M_1 &= \langle x^2, x^5, x^{11} + \frac{11}{4}x^8 \rangle, \\ M_2 &= \langle x^3, x^6, x^9 \rangle. \end{aligned}$$

The quartic has monodromy S_4 , whose reduced permutation representation is irreducible. This forces all three non-trace quartic channels. A fibre-trace lemma then removes the additional quartic trace channel after composition, while a modulo-three grading fuses the surviving channels into three explicit blocks modulo twelve.

For the third result, let

$$\Delta_C(t, \epsilon; \mathbf{a}) = \sum_{\mu \geq 1} \epsilon^\mu M_\mu(t; \mathbf{a})$$

be the displacement of a zero-cycle under a generic polynomial perturbation $g_{\mathbf{a}}$ of arbitrary fixed degree. Parameter content is the ideal generated by the polynomial coefficients of an algebraic function when expanded in any complex basis of its finite coefficient span. If $\mathcal{B}_k$ is generated by the contents of $M_1, \dots, M_k$, then

$$\mathcal{B}_\infty = \mathcal{B}_{d-1}.$$

Differentiation is injective on the vector space of zero-cycle polynomial integrals: a differentiated integral in the kernel would be a single-valued polynomial in t , and averaging its infinity-monodromy orbit annihilates the zero-cycle. Hence differentiation preserves parameter content exactly. Cayley–Hamilton for multiplication by g in the rank- d algebra $R[t, x]/(f - t)$ then shows that no power moment, and therefore no Melnikov coefficient, introduces a generator beyond order $d - 1$. In particular, $b_{\text{coeff}}(m) \leq m - 1$.

As a supporting bridge, the first dependence among

$$[A \, dx], [PA \, dx], [P^2 A \, dx], \dots$$

in the twisted quotient produces an operator annihilating every rapid-decay period. Using the reconstructed theorem and Hien’s perfect pairing between algebraic de Rham cohomology and rapid-decay homology, its minimal complete-system order is

$$|\Sigma(P; G)| = \dim \text{span}_{\text{red}} \{G(x_i)\}.$$

For a ray-cycle whose Fourier transform touches the channel set R , the individual order lies between $|R|$ and the total mass of the monodromy blocks meeting R . Full contact or differential-module irreducibility makes the appropriate bound sharp. This is a qualified individual-cycle statement, not the blanket channel-count formula rejected below and not the third headline claim.

Prior-art boundary. Puiseux expansions of $Q(P_i^{-1})$, Fourier support separation, monodromy subspaces and zero-dimensional Abelian integrals are established in Pakovich–Muzychuk [1], Alvarez Sanchez–Bravo Trinidad–Mardesic [2], and Gavrilov–Pakovich [3]. Pakovich’s fixed-segment Chebyshev moment theorem [13] gives a Chebyshev-derivative basis for the amplitudes whose moments all vanish; it is a close structural antecedent but does not state the cyclic rank of every amplitude for the complete rapid-decay period system. Individual Chebyshev/Lucas amplitudes in exponential integrals and their Bessel or Airy–Lucas equations occur in [4,5].

Pakovich–Muzychuk [1] also classify invariant subspaces of a polynomial monodromy group containing a full cycle. The representation-theoretic block decomposition in Section 2.1 is therefore not claimed as new. The residual steps used here are the descent of central projectors to polynomial $\mathbf{C}[P]$ -lattices, realization of every nonzero lattice component as the full inverse-support character block, and the resulting all-amplitude rank formulas.

The exact phase $(x^4 + x)^3$ is already identified as a Ritt collision with monodromy $S_4 \times C_3$ by Behajaina–Konig–Neftin [6, Theorems 2.2(1) and 2.3(1)]. Our residual candidate contribution

in Problem B is therefore the amplitude-module/support classification and all-amplitude rank formula, not the collision or its monodromy group.

Minimal hyperexponential telescopers from first reduced dependence are prior art [7], as are order-aware and reduction-based creative-telescoping methods [8,9]. Rapid-decay period representations for exponential Gauss–Manin systems are also established [10], and Hien proves the required perfect period pairing [12, Theorem 5.2]. The support-to-order bridge is presented only as the Atlas-specific identity and an exact benchmark.

Bravo–Mardešić–Novikov–Pontigo–Herrera [17] ask for a degree-uniform zero-cycle Bautin bound. Under the ordinary Taylor-coefficient reading of their ideal chain, Proposition 6.1 identifies that coefficient ideal with the parameter-content ideal used here and Problem C gives $b_{\text{coeff}}(m) \leq m - 1$. If [17] intends its terse $b(m)$ in this sense, then $b(m) = b_{\text{coeff}}(m)$. Batenkov–Yomdin [18] define ordinary Bautin ideals by Taylor coefficients and prove a general Bautin-ideal theorem for polynomial recurrences, while Briskin–Yomdin [19] treat algebraic families of analytic functions. These are close antecedents. The residual step here is to prove that the order-dependent differentiation taking power moments to Melnikov coefficients preserves parameter content on a zero-cycle. The source [17] does not specify a different localization, radicalization, or analytic-parameter ideal; none of those alternatives is claimed here. The bounded search found no exact statement of the $d - 1$ bound, but does not establish priority.

The broader block theorem, the cycle-by-cycle order question and the binomial Laurent specialization were also tested during development. The manuscript gives self-contained arguments for its principal retained candidate conclusions. The unrestricted exact cycle formula fails by the counterexample in Section 8.1; only the proved standard-ray block bound is retained. The former general Dickson individual-cycle equality was removed because its scalar differential-module derivation was incomplete. The binomial result appears in Appendix A as a direct Pakovich-based corollary, not as a fourth novelty claim.

Theorem-level dependency and prior-art map.

Candidate statement	Imported or closest result	Residual statement here	Assurance ceiling
Theorem 2.1	Atlas [14]; invariant-subspace classification [1, Theorem 3.1]; zero-cycle composition structure [2, Theorem 2.2]	completed formal-moment isomorphism and equality of cyclic rank, inverse support and reduced root-value span	same-programme candidate proof
Theorem A	Pakovich’s main Chebyshev moment theorem [13]; Airy–Lucas examples [4, Section 3; 5, Sections 5.2–5.3]	all-amplitude free module and exact rank law $2N + \varepsilon$	candidate proof plus bounded exact checks
Theorem B	exceptional monodromy $S_4 \times C_3$ [6]	explicit five-summand amplitude module, fibre-trace cancellation and all-amplitude rank formula	candidate proof plus bounded exact checks

Candidate statement	Imported or closest result	Residual statement here	Assurance ceiling
Support-to-order bridge	first-dependence telescoping [7–9]; rapid-decay realization [10]; perfect pairing [12, Theorem 5.2]	Atlas support predicts the complete-system minimal order; corrected standard-ray bounds	supporting bridge, not a headline novelty claim
Theorem C	Problem 8.4 of [17]; ordinary coefficient Bautin ideals and recurrences [18]	differentiation preserves parameter content on zero-cycle integrals and yields the explicit cutoff $d - 1$	candidate theorem under the coefficient convention of Proposition 6.1
Appendix A	Pakovich’s Laurent theorem [11, Theorem 6.5]	elementary binomial specialization and module-rank count	contextual corollary only

The dependency is linear only at the first two headline results: Theorems A and B use Theorem 2.1 to convert their polynomial-module classifications into cyclic-rank formulas. The complete-system bridge additionally uses Hien’s pairing. Theorem C and Appendix A do not depend on Theorem 2.1.

THE RECONSTRUCTED SUPPORT THEOREM

Let P have degree d , let $\xi(t)$ be the normalized inverse branch at infinity satisfying $P(\xi(t)) = t^d$, and let $G' = A$. Write $\Sigma(P; G) \subset \mathbf{Z}/d\mathbf{Z}$ for the nonzero, non-trace residues represented in the Laurent expansion of $G(\xi(t))$.

For a generic specialized regular value y , meaning a value outside the finite excluded set in Lemma 2.2 below, let $P^{-1}(y) = \{x_1, \dots, x_d\}$. Define the reduced root-value span to be the complex span of the monodromy orbit of

$$(G(x_1), \dots, G(x_d)) \quad \text{modulo the constant vector.}$$

Define $q(P, G' dx)$ to be the dimension over $\mathbf{C}(s)$ of the cyclic span of $[G' dx]$ under the connection induced by multiplication by P in the twisted quotient below.

Theorem 2.1 (polynomial-amplitude support identity). For every polynomial phase P and polynomial primitive G ,

$$(H) \quad q(P, G' dx) = |\Sigma(P; G)| = \dim \text{span}_{\text{red}} \{G(x_i)\}.$$

Proof. Put $K = \mathbf{C}(s)$ and

$$\mathcal{H}_P = K[x] dx / (\partial_x + sP') K[x] dx.$$

After affine normalization, P is monic and centred, and $\mathcal{H}_P$ has the basis $[dx], \dots, [x^{d-2} dx]$. Over $K_0 = \mathbf{C}((s))$, introduce rank-one modules $\mathcal{K}_r = K_0 e_r$, $1 \leq r \leq d - 1$, with

$$\partial_s e_r = -\frac{r}{d} s^{-1} e_r.$$

For every $n \in \mathbf{Z}$, define

$$\mu_{r,0} = 1, \quad \mu_{r,n+1} = -\frac{r+nd}{d}\mu_{r,n},$$

forward for $n \geq 0$ and backward for $n < 0$; explicitly, the same equation uniquely gives

$$\mu_{r,n-1} = -\frac{d}{r+(n-1)d}\mu_{r,n}.$$

and on the space of bounded-above Laurent one-forms in residue class $r-1$ send a monomial to

$$(2.1) \quad t^{r+nd-1}dt \mapsto \mu_{r,n}s^{-n}e_r.$$

Every $\mu_{r,n}$ is nonzero. A bounded-above t -series maps to an s -Laurent series of lower-bounded valuation; after scalar extension to K_0 , each fixed s -coefficient receives only finitely many contributions. Thus the transform uses no uncontrolled double Laurent series. Since $P(\xi) = t^d$, formal integration by parts shows that (2.1) kills the pullback of $(\partial_x + sP')K[x]dx$, and differentiation in s corresponds to multiplication by P . The map therefore descends to a differential-module morphism

$$(2.2) \quad \mathcal{H}_P \otimes_K K_0 \longrightarrow \bigoplus_{r=1}^{d-1} \mathcal{K}_r.$$

On $x^j dx$, $0 \leq j \leq d-2$, one has $\xi^j \xi' = t^j + O(t^{j-2})$. The matrix of (2.2) lies in $\text{Mat}_{d-1}(\mathbf{C}[[s]])$ and has triangular constant term with diagonal one. It is therefore an isomorphism.

Now

$$d(G(\xi)) = \sum_m m c_m t^{m-1} dt.$$

The r -component is nonzero exactly when $r \in \Sigma(P; G)$, because its distinct Laurent coefficients occupy distinct powers of s and no factor $r+nd$ vanishes. If n_r is the largest active index in residue r , let $b_r \neq 0$ be the coefficient of its leading $s^{-n_r}e_r$ -term and put $\alpha_r = -n_r - r/d$. The first $m = |\Sigma(P; G)|$ derivative columns have leading coefficient matrix

$$((\alpha_r)_{\underline{j}})_{r \in \Sigma(P; G), 0 \leq j < m}.$$

Its determinant is the Vandermonde $\prod_{r < r'} (\alpha_{r'} - \alpha_r) \neq 0$. Since $1 \leq r \leq d-1$, $\alpha_r = -n_r - r/d$ is nonintegral, so every individual falling factorial here is nonzero. The point of retaining $(\alpha_r)_{\underline{j}} + O(s)$ is that the leading determinant is the whole falling-factorial matrix, not an unsupported entrywise factorization. More explicitly, the direct sum of the active Kummer lines is an m -dimensional differential submodule containing the seed, so the cyclic rank is at most m . In the Wronskian, factor $b_r s^{-n_r}$ from row r and s^{-j} from derivative column j . The remaining constant matrix is exactly $((\alpha_r)_{\underline{j}})$, whose Vandermonde determinant is nonzero; hence the first m derivatives are independent and the rank is at least m . Finally, rank of a finite matrix over $K = \mathbf{C}(s)$ is unchanged after the faithful field extension $K \subset \mathbf{C}((s))$, so this is the rank of the original cyclic span, not only its completion.

To identify this formal support with the full monodromy-orbit quantity, label the inverse roots near infinity by $x_j(t) = \xi(\zeta_d^j t)$, $0 \leq j < d$, and put

$$Y_r(t) = \frac{1}{d} \sum_{j=0}^{d-1} \zeta_d^{-jr} G(x_j(t)) = \sum_{m \equiv r \pmod{d}} c_m t^m.$$

Lemma 2.2 (equivariant span and generic specialization). Let $\mathcal{L}$ be a splitting field of $P(X) - y$ over $\mathbf{C}(y)$, let Γ be its geometric monodromy group, and set

$$\bar{g}_j = G(x_j) - \frac{1}{d} \sum_{i=0}^{d-1} G(x_i), \quad W_G = \text{span}_{\mathbf{C}}\{\bar{g}_0, \dots, \bar{g}_{d-1}\} \subset \mathcal{L}.$$

Then

$$(2.3) \quad \dim_{\mathbf{C}} W_G = |\Sigma(P; G)|.$$

Moreover, there is a finite algebraic set $Z_G \subset \mathbf{A}_y^1$, containing the critical values of P , such that for every $y_0 \notin Z_G$, and for any simultaneous determination of the roots at y_0 , the full Γ -orbit of the reduced vector $(\bar{g}_0(y_0), \dots, \bar{g}_{d-1}(y_0))$ spans a complex vector space of dimension $\dim W_G$.

Proof. Let

$$E = \{(a_0, \dots, a_{d-1}) \in \mathbf{C}^d : \sum_j a_j = 0\}$$

and define the Γ -equivariant map

$$\Phi_G : E \longrightarrow \mathcal{L}, \quad (a_j) \longmapsto \sum_j a_j G(x_j).$$

Its image is W_G . The non-trace Fourier vectors form a basis of E , and their images are the functions dY_r , $1 \leq r < d$. The nonzero Y_r are constant-linearly independent because their Laurent supports occupy disjoint residue classes. Hence the rank of Φ_G is exactly $|\Sigma(P; G)|$, proving (2.3).

The inertia element at infinity is a d -cycle, so it has simple spectrum on $\mathbf{C}^d$. Consequently the Γ -permutation module is multiplicity-free: if one irreducible constituent occurred twice, every character occurring in its restriction to the inertia cycle would occur at least twice, contradicting simple spectrum. Its quotient W_G is therefore also multiplicity-free. Write $W_G = \bigoplus_{\nu} W_{\nu}$ as its sum of pairwise nonisomorphic irreducible constituents. For every nonzero W_{ν} , choose a nonzero algebraic function $h_{\nu} \in W_{\nu}$. Enlarge the critical-value set by the projections to the y -line of the finitely many zeros and poles of the chosen h_{ν} on the compactified finite cover, and call the resulting finite set Z_G .

For $y_0 \notin Z_G$, evaluation on the chosen simultaneous determination defines $\lambda_{y_0} \in W_G^*$, and its restriction to every W_{ν} is nonzero. Since W_G^* is multiplicity-free, the Γ -submodule generated by λ_{y_0} is all of W_G^* : central idempotents first isolate its nonzero component in each irreducible summand, and irreducibility then generates that summand. The dual injection

$$\Phi_G^* : W_G^* \hookrightarrow E^*$$

is Γ -equivariant. Under the standard invariant identification $E^* \simeq E$, the vector $\Phi_G^*(\lambda_{y_0})$ is precisely $(\bar{g}_0(y_0), \dots, \bar{g}_{d-1}(y_0))$. Its full monodromy-orbit span therefore has dimension $\dim W_G$. A different simultaneous determination applies an element of Γ and does not change that dimension. $\square$

Combining Lemma 2.2 with the cyclic-rank calculation above proves (H). $\square$

Lemma 2.3 (standard-ray flat family and Fourier pairing). Fix a simply connected sector

$$\mathfrak{S} = \{s \neq 0 : |\arg s - \theta_0| < \varepsilon\}, \quad 0 < \varepsilon < \pi/2,$$

and continuous branches of $\arg s$ and $s^{1/d}$ on it. For $s = |s|e^{i\theta}$, set

$$\phi_k(s) = \frac{\pi - \theta + 2\pi k}{d}, \quad \ell_k(s) = e^{i\phi_k(s)}[0, \infty),$$

and orient $\Gamma_k(s)$ from infinity to the origin on $\ell_k(s)$, then from the origin to infinity on $\ell_{k+1}(s)$. These contours define a flat family of rapid-decay classes on $\mathfrak{S}$, with

$$\sum_{k=0}^{d-1} [\Gamma_k] = 0, \quad [\Gamma_0], \dots, [\Gamma_{d-2}] \text{ form a basis.}$$

For $r = j + 1$, as $s \rightarrow 0$ within any compact subsector of $\mathfrak{S}$,

$$(2.4) \quad s^{r/d} \int_{\Gamma_k} x^j e^{sP(x)} dx \longrightarrow \frac{\Gamma(r/d)}{d} e^{\pi i r/d} (\zeta_d^r - 1) \zeta_d^{kr}.$$

The resulting $(d-1)$ -by- $(d-1)$ leading matrix has full rank. Under the formal-moment isomorphism (2.2), its Fourier diagonalization gives a nondegenerate pairing between the r -th Kummer line and the dual ray-cycle Fourier mode. Thus a weighted standard ray-cycle has a nonzero r -projection exactly when its Fourier coefficient and the r -component of $G(\xi)$ are both nonzero.

Proof. Along $\ell_k(s)$, the leading term satisfies $sx^d = -|s||x|^d$. The ends therefore stay in consecutive rapid-decay sectors and vary continuously with s ; contour deformation inside those sectors supplies the flat family. For a large disk, rapid-decay relative homology retracts to the reduced zero-dimensional homology of its d marked decay intervals. The boundary vector of Γ_k is $e_{k+1} - e_k$, so the displayed sum is its sole relation and any consecutive $d-1$ classes form a basis.

After the scaling $u = s^{1/d}x$, the moving rays become the fixed rays $\arg u = (\pi + 2\pi k)/d$, and centredness gives

$$sP(s^{-1/d}u) = u^d + \sum_{\ell=0}^{d-2} p_\ell s^{1-\ell/d} u^\ell.$$

On every compact subsector, Young's inequality gives constants $c, C > 0$, independent of sufficiently small s , such that the real part of the right side is at most $-c|u|^d + C$ on all the scaled rays. Thus $C(1 + |u|)^j e^{-c|u|^d}$ is an integrable majorant, differentiation under the integral is valid for polynomial amplitudes, and dominated convergence reduces the limit to the two adjacent monomial rays for e^{u^d} . Direct Gamma integration, with the stated incoming and outgoing orientations, gives (2.4).

For $0 \leq k \leq d-2$ and $1 \leq r \leq d-1$, the matrix in (2.4) is a nonzero diagonal matrix times (ζ_d^{kr}) . Its determinant is the Vandermonde determinant on $\zeta_d, \dots, \zeta_d^{d-1}$, hence is nonzero. Continuation once around $s = 0$ changes ϕ_k to ϕ_{k-1} , so it sends $[\Gamma_k]$ to $[\Gamma_{k-1}]$. The ray Fourier modes consequently have the same distinct characters as the Kummer lines in (2.2), and the nonzero diagonal leading matrix identifies the dual characters. This proves the pairing and channel-contact assertions. $\square$

Lemma 2.3 is the channel-contact input used in Section 5. It requires neither a Borel-summability claim nor a statement about arbitrary contour systems.

General monodromy blocks. Let $\Gamma \leq S_d$ be the monodromy group and let $c \in \Gamma$ be the inertia d -cycle at infinity. The restriction of the permutation module to $\langle c \rangle$ is the regular cyclic representation, so all c -characters occur once. An irreducible Γ -constituent cannot occur with multiplicity greater than one, because every character in its cyclic restriction would then occur at least twice. Hence

$$\mathbf{C}^d = U_0 \oplus U_1 \oplus \cdots \oplus U_k$$

is multiplicity-free. Each U_i is the span of a unique block $B_i \subset \mathbf{Z}/d\mathbf{Z}$ of Fourier characters, with $B_0 = \{0\}$.

Let $e_i \in \mathbf{C}[\Gamma]$ be the central projector onto U_i . Applied to the root-value vector of a polynomial G , its first coordinate is fixed by the root stabilizer and therefore belongs to $\mathbf{C}(x)$. Every generic root is integral over $\mathbf{C}[y]$, so every polynomial root value and every constant linear combination of such values is integral over $\mathbf{C}[y]$. The integral closure of $\mathbf{C}[y]$ in $\mathbf{C}(x)$, with $y = P(x)$, is $\mathbf{C}[x]$; hence the projected coordinate is a polynomial. Thus e_i descends to a $\mathbf{C}[P]$ -linear projector $T_i : \mathbf{C}[x] \rightarrow \mathbf{C}[x]$.

Writing $\mathcal{M}_i = \text{im } T_i$ gives projective direct summands of the free $\mathbf{C}[P]$ -module $\mathbf{C}[x]$. They are free because $\mathbf{C}[P]$ is a PID, and

$$(2.5) \quad \mathbf{C}[x] = \bigoplus_{i=0}^k \mathcal{M}_i, \quad \mathcal{M}_0 = \mathbf{C}[P], \quad \text{rank}_{\mathbf{C}[P]} \mathcal{M}_i = |B_i|.$$

For completeness, extend scalars from $\mathbf{C}(P)$ to a splitting field of $P(x) - y$. Generic evaluation identifies $\mathbf{C}(x) \otimes_{\mathbf{C}(P)} \Omega$ with Ω^d , and the extended image of T_i is $U_i \otimes \Omega$. Rank is preserved by this faithfully flat field extension, so $\text{rank}_{\mathbf{C}[P]} \mathcal{M}_i = \dim U_i = |B_i|$, as asserted in (2.5).

If $0 \neq G_i \in \mathcal{M}_i$, its monodromy orbit spans the irreducible module U_i , while its inverse support is contained in B_i . Theorem 2.1 equates the orbit-span dimension and support cardinality, so $\Sigma(P; G_i) = B_i$. Therefore

$$(2.6) \quad \Sigma(P; G) = \bigcup_{\substack{i \geq 0 \\ T_i G \neq 0}} B_i.$$

In particular, the attainable supports are exactly the unions of non-trace blocks. For arbitrary S , the module of primitives with support contained in S is the direct sum of the trace module and those $\mathcal{M}_i$ whose complete block B_i lies in S ; its rank is $1 + \sum_{B_i \subset S} |B_i|$.

The central-projector descent is important. Dimension counting after extension to a formal Laurent field alone would not show that the resulting subspace has enough polynomial lattice points.

DICKSON AMPLITUDE MODULES

Let $a \in \mathbf{C}^\times$, $d \geq 2$, and define Dickson polynomials by

$$D_j(u + a/u, a) = u^j + a^j u^{-j}.$$

Put $P = D_d(x, a)$.

The module decomposition. Every $G \in \mathbf{C}[x]$ has a unique expansion

$$(3.1) \quad G = H_0(P) + \sum_{1 \leq j < d/2} (H_j(P)D_j + K_j(P)D_{d-j}) + \mathbf{1}_{2|d}H_{d/2}(P)D_{d/2}.$$

Indeed, $D_j(x, a)$ is monic of degree j . Therefore

$$1, D_1, \dots, D_{d-1}$$

is a triangular change of basis from $1, x, \dots, x^{d-1}$, which is a basis of the free $\mathbf{C}[P]$ -module $\mathbf{C}[x]$.

Let $N(G)$ be the number of nonzero paired summands in (3.1), and let $\varepsilon(G) = 1$ exactly when the midpoint term is present.

Quadratic obstruction to channel loss. Choose w and u near infinity such that

$$w + \frac{a^d}{w} = t^d, \quad u^d = w, \quad \xi(t) = u + \frac{a}{u}.$$

Then $D_d(\xi(t), a) = t^d$. With $Y = t^d$, the element w satisfies

$$(3.2) \quad w^2 - Yw + a^d = 0.$$

Its discriminant $Y^2 - 4a^d$ has two simple zeros and is not a square in $\mathbf{C}(Y)$. Hence $w \notin \mathbf{C}(Y)$.

Fix $1 \leq j < d/2$, abbreviate $F = H_j$ and $K = K_j$, and evaluate its paired summand on $\xi(t)$:

$$(3.3) \quad \begin{aligned} F(P)D_j + K(P)D_{d-j} &= w^j \left(F(Y) + \frac{a^{d-j}K(Y)}{w} \right) \\ &\quad + u^{-j} \left(a^j F(Y) + K(Y)w \right). \end{aligned}$$

The two lines occupy residues j and $d-j$ modulo d . If the first coefficient vanished, then either $F = K = 0$, or $w = -a^{d-j}K/F \in \mathbf{C}(Y)$, a contradiction. The same argument applies to the second coefficient. Thus every nonzero pair activates exactly two channels. Different unordered pairs have disjoint residues, the midpoint has one residue, and $H_0(P)$ is trace-only.

Theorem A.

$$(3.4) \quad \boxed{q(D_d, G' dx) = 2N(G) + \varepsilon(G).}$$

For odd d , the attainable ranks are $0, 2, \dots, d-1$. For even d , every rank $0, 1, \dots, d-1$ occurs. Rank zero is equivalent to $G \in \mathbf{C}[P]$.

THE EXCEPTIONAL RITT AMPLITUDE MODULE

Put

$$Q = x^4 + x, \quad P = Q^3.$$

The phase has the two length-two decompositions

$$(4.0) \quad P = X^3 \circ (X^4 + X) = (X(X + 1)^3) \circ X^3.$$

The collision and its $S_4 \times C_3$ monodromy are imported from [6, Theorems 2.2(1) and 2.3(1)]. The residual candidate developed here is the explicit amplitude-module/support classification.

Define M_0, M_1, M_2 as in Section 1.1.

Direct decomposition. The twelve polynomials

$$1, Q, Q^2, \quad \text{and the nine displayed generators of } M_0, M_1, M_2$$

are monic with distinct leading degrees $0, \dots, 11$. Since P is monic of degree twelve, they form a $\mathbf{C}[P]$ -basis. Consequently, every G decomposes uniquely as

$$(4.1) \quad G = A_0(P) + A_1(P)Q + A_2(P)Q^2 + m_0 + m_1 + m_2.$$

In particular, $M_i \cap \mathbf{C}[Q] = \{0\}$.

The quartic root module. The critical points of Q are the three roots of $4x^3 + 1 = 0$. At a critical point c ,

$$Q(c) = \frac{3}{4}c, \quad Q''(c) = 12c^2 \neq 0.$$

Thus the finite critical values are distinct and their branch cycles are transpositions. The degree-four cover is connected, so the graph defined by these transpositions is connected. Edge transpositions of a connected graph generate S_4 . Hence

$$\text{Mon}(Q) = S_4.$$

The reduced permutation representation is the irreducible three-dimensional standard representation. If $R \notin \mathbf{C}[Q]$, its reduced root-value vector is nonzero, so the span of its continuations is the entire standard module. The equivariant generic-specialization lemma in the proof of Theorem 2.1 applies to Q : outside its finite excluded set, the full orbit and conjugate-function spans both have dimension three. Theorem 2.1 then says that the inverse series of R activates all three nonzero residues modulo four.

Fusion to degree twelve. The normalized inverse of Q has the form

$$(4.2) \quad \eta(t) = t\phi(t^{-3}), \quad \phi(z)^4 + z\phi(z) = 1, \quad \phi(0) = 1.$$

All monomial degrees in M_0, M_1, M_2 are respectively congruent to $1, 2, 0$ modulo three. Multiplication by P preserves those classes. The quartic trace channel requires a separate exclusion because it becomes a non-trace channel after composition with the outer cube.

For $R \in \mathbf{C}[x]$, put

$$\mathrm{tr}_Q R(y) = \sum_{Q(u)=y} R(u).$$

Writing $p_j(y) = \mathrm{tr}_Q(x^j)$, Newton's identities for $x^4 + x - y$ give

$$\begin{aligned} p_1 = p_2 = p_5 = 0, \quad p_3 = -3, \quad p_6 = 3, \quad p_9 = -3, \\ p_4 = 4y, \quad p_7 = -7y, \quad p_{10} = 10y, \\ p_8 = 4y^2, \quad p_{11} = -11y^2. \end{aligned}$$

Thus the displayed generators of M_0 and M_1 are Q -trace-free, whereas those of M_2 have constant traces $-3, 3, -3$. Under arbitrary $\mathbf{C}[P]$ -coefficients the first two traces remain zero, while the third remains supported only in powers of y^3 , because

$$\mathrm{tr}_Q(F(P)R) = F(y^3) \mathrm{tr}_Q R.$$

Moreover,

$$\frac{1}{4} \mathrm{tr}_Q R(t^4) = \frac{1}{4} \sum_{j=0}^3 R(\eta(i^j t))$$

is exactly the part of $R(\eta(t))$ with exponent divisible by four. It vanishes on M_0, M_1 ; on M_2 it lies in powers of t^{12} , the trace class modulo twelve. Every nonzero m_i also has all three nonzero quartic channels by Section 4.2. The trace filter, the fixed modulo-three class and the Chinese remainder theorem therefore give

$$(4.3) \quad \begin{array}{l|l} M_0 & \{1, 7, 10\} \\ M_1 & \{2, 5, 11\} \\ M_2 & \{3, 6, 9\}. \end{array}$$

Meanwhile $Q(\eta(t)) = t^4$, so the two non-trace composition channels are $\{4\}$ and $\{8\}$. These five blocks partition the eleven nonzero residues modulo twelve.

Theorem B. The unique expansion (4.1) satisfies

$$(4.4) \quad \boxed{q(P, G' dx) = \mathbf{1}_{A_1 \neq 0} + \mathbf{1}_{A_2 \neq 0} + 3 \sum_{i=0}^2 \mathbf{1}_{m_i \neq 0}.$$

Every rank from zero through eleven occurs. A nonzero element of one fused module has rank exactly three, at every coefficient degree.

SUPPORT-PREDICTED WEIGHTED PERIOD OPERATORS

Let $P, A \in \mathbf{C}[x]$, $\deg P = d \geq 2$, and $G' = A$. Define

$$\mathcal{H}_P = \frac{\mathbf{C}(s)[x] dx}{(\partial_x + sP'(x))\mathbf{C}(s)[x] dx}$$

and $v_j = [P^j A dx]$. Let q be the least index for which $v_0, \dots, v_q$ are dependent.

Exact reduction. Write

$$P'(x) = px^{d-1} + L(x), \quad \deg L < d-1.$$

The twisted derivative of x^k gives

$$(5.1) \quad [x^{k+d-1} dx] = -\frac{k}{sp}[x^{k-1} dx] - \frac{1}{p}[x^k L(x) dx].$$

Every replacement lowers degree, so reduction terminates in the basis

$$dx, x dx, \dots, x^{d-2} dx.$$

The representative is unique: a nonzero twisted derivative of a polynomial f has leading degree $\deg f + d - 1$, and therefore cannot be a nonzero polynomial of degree below $d - 1$.

Operator and exact witness. The first dependence has a unique monic rational form

$$v_q + \sum_{j < q} c_j(s) v_j = 0.$$

Pairing with a flat rapid-decay cycle and differentiating under the integral produces

$$\left(D_s^q + \sum_{j < q} c_j(s) D_s^j \right) I_{\Gamma, A} = 0.$$

For rational inputs, write the left operator as $L = \sum_{j=0}^q p_j(s) D_s^j$. Take the least common multiple in $\mathbf{Q}[s]$ of the denominators of the monic rational coefficients, divide the resulting polynomial coefficients by their greatest common polynomial factor, clear the remaining scalar denominators, divide by the positive greatest common divisor of all integer coefficients, and choose the sign so that the leading coefficient of $p_q(s)$ is positive. This gives a unique primitive representative in $\mathbf{Z}[s]\langle D_s \rangle$; the Ore coefficients are written on the left, and the only units removed are nonzero rational scalars and common polynomial factors.

The reduction also reconstructs $C(x, s) \in \mathbf{Q}(s)[x]$ satisfying

$$(5.2) \quad \sum_j c_j(s) P(x)^j A(x) = \partial_x C(x, s) + sP'(x)C(x, s).$$

Equation (5.2) is a directly checkable annihilation certificate.

Hien's perfect-pairing theorem [12, Theorem 5.2] is stated for flat algebraic connections in all dimensions, so it applies in particular to the rank-one exponential connection on $\mathbf{A}^1$. This scope and theorem number were checked directly in the public arXiv source. After specializing

rational coefficient functions at a generic nonsingular value of s , an operator of order $r < q$ annihilating every period would pair a nonzero class in the span of $v_0, \dots, v_r$ to zero against every rapid-decay homology class. This contradicts perfectness.

Support-to-order theorem.

$$(5.3) \quad \boxed{\text{ord}_{\min, \text{complete}}(P, A) = q = |\Sigma(P; G)| = \dim \text{span}_{\text{red}}\{G(x_i)\}.$$

Here minimality refers to one operator annihilating the complete period system. A specified individual cycle may satisfy a lower-order equation. No generic-individual-period claim is made. If $q = 0$, then $[A dx] = 0$ and every period is identically zero.

A corrected cycle-specific refinement. Use the standard ray-cycle basis at infinity. For a zero-trace coefficient vector n , the Fourier–Gamma realization (2.4) and the Kummer decomposition (2.2) give the sectorial monodromy-channel decomposition

$$I_n(s) = \sum_{r \in \Sigma(P; G)} \hat{n}(r) \Lambda_r(s),$$

where $\Lambda_r \neq 0$ has formal-monodromy eigenvalue $\exp(-2\pi i r/d)$. Put

$$R = \text{supp}(\hat{n}) \cap \Sigma(P; G)$$

and let $B_1, \dots, B_t$ be the monodromy blocks meeting R . If an order- r rational scalar operator annihilates I_n , its completed cyclic differential module at $s = 0$ has rank r and is stable under formal monodromy. Lagrange interpolation in that monodromy projects the formal expansion of I_n onto its nonzero components with distinct eigenvalues; those projections are independent, so $r \geq |R|$. The central-projector decomposition supplies a complete-system operator of order $|B_i|$ for each touched block. Hence

$$(5.4) \quad \boxed{|R| \leq \text{ord}_{\min}(I_n) \leq \sum_{i=1}^t |B_i| \leq q(P, G' dx).$$

If every touched block is fully touched, the first two quantities in (5.4) coincide. No general equality with the block-mass upper bound is claimed; such an equality requires a separate irreducibility analysis of the touched differential modules.

THE UNIFORM ZERO-CYCLE BAUTIN BOUND

Let $f \in \mathbf{C}[x]$ be monic of degree $d \geq 2$, let

$$C(t) = \sum_{j=1}^d n_j x_j(t), \quad \sum_j n_j = 0,$$

be any zero-cycle of f , and fix an arbitrary perturbation degree n :

$$g_a(x) = \sum_{r=0}^n a_r x^r, \quad R = \mathbf{C}[a_0, \dots, a_n].$$

Continue the original roots by $f(z_j) + \epsilon g_a(z_j) = t$. With $C_\epsilon(t) = \sum_j n_j z_j(t, \epsilon)$, define the displacement exactly as in [17]:

$$\Delta_C(t, \epsilon; \mathbf{a}) = \int_{C_\epsilon(t)} f = -\epsilon \sum_j n_j g_{\mathbf{a}}(z_j(t, \epsilon)),$$

and write

$$\Delta_C(t, \epsilon; \mathbf{a}) = \sum_{\mu \geq 1} \epsilon^\mu M_\mu(t; \mathbf{a}).$$

If $n > d$, additional roots of the perturbed equation tend to infinity and receive zero weights, exactly as in [17]. Fix a simply connected regular t -domain and let $\mathcal{K}$ be the algebra of algebraic-function germs there. For $F \in R \otimes_{\mathbf{C}} \mathcal{K}$, define $\text{cont}_R(F)$ to be the ideal generated by the R -coordinates of F in any complex basis of its finite coefficient span. A complex change of basis leaves the ideal unchanged. Set

$$\mathcal{B}_k = \sum_{\mu=1}^k \text{cont}_R(M_\mu), \quad \mathcal{B}_\infty = \sum_{\mu \geq 1} \text{cont}_R(M_\mu).$$

These are ordinary coefficient ideals, not radicals or only centre loci.

Proposition 6.1 (content equals the local Taylor-coefficient ideal). For a regular base point t_0 , write

$$F(\mathbf{a}, t) = \sum_{k \geq 0} \tau_k(\mathbf{a})(t - t_0)^k.$$

Then

$$(6.1) \quad \text{cont}_R(F) = (\tau_0, \tau_1, \dots) \subset R.$$

Indeed, if $F = \sum_{\ell=1}^s b_\ell(\mathbf{a})v_\ell(t)$ in a complex basis of its coefficient span, every τ_k is a complex linear combination of the b_ℓ . Conversely, linear independence of the analytic germs v_ℓ implies that their infinite Taylor-coefficient vectors are linearly independent. Some finite s -by- s jet submatrix is therefore invertible, expressing every b_ℓ as a complex linear combination of finitely many τ_k . This proves (6.1), independently of the chosen regular point or determination because analytic continuation and change of basis act invertibly on the finite germ span. Thus the content ideal is exactly the ordinary Taylor-coefficient Bautin ideal of [18].

For clarity, define the source-style coefficient ideal at a regular point by

$$\mathcal{I}_k^{\text{coeff}}(t_0) = (\text{all Taylor coefficients of all local determinations of } M_1, \dots, M_k) \subset R.$$

Proposition 6.1 gives the equality of ideals $\mathcal{I}_k^{\text{coeff}}(t_0) = \mathcal{B}_k$, not merely equality of radicals or centre loci. Analytic continuation and a change of regular base point act by invertible complex changes on the finite germ spans, so the equality is branch- and base-point-independent. Localizing the parameter ring preserves it: $S^{-1}\mathcal{I}_k^{\text{coeff}} = S^{-1}\mathcal{B}_k$ for every multiplicative set $S \subset R$. Equality therefore also implies equality of radicals and of affine centre loci for this coefficient convention. No equivalence is asserted with an ideal formed inside a larger analytic-function ring, with radicalization before contraction to R , or at a singular t -value.

All-order Melnikov identity. Put $J_\mu = \int_C g_a^\mu$. Lagrange–Bürmann inversion on each continued root branch gives

$$(6.2) \quad M_\mu(t; \mathbf{a}) = \frac{(-1)^\mu}{\mu!} \left(\frac{d}{dt} \right)^{\mu-1} J_\mu(t; \mathbf{a}), \quad \mu \geq 1.$$

Indeed, if $G_j(u) = g(x_j(u))$ and $u_j + \epsilon G_j(u_j) = t$, Lagrange–Bürmann gives

$$G_j(u_j(t)) = \sum_{\nu \geq 0} \frac{(-\epsilon)^\nu}{(\nu+1)!} D_t^\nu (G_j^{\nu+1})(t).$$

Multiplication by $-\epsilon n_j$ and summation gives (6.2). No comparison between $\deg f$ and $\deg g$ enters this branchwise identity.

Monodromy injectivity. Let

$$\mathcal{V}_C = \left\{ \int_C h : h \in \mathbf{C}[x] \right\}.$$

Lemma 6.2. For every $q \geq 1$, D_t^q is injective on $\mathcal{V}_C$.

Proof. If $F = \int_C h$ and $D_t^q F = 0$, then locally F is a polynomial in t of degree less than q , hence is single-valued. Monodromy τ around infinity is a d -cycle on the branches of f . Since C has total weight zero,

$$\frac{1}{d} \sum_{k=0}^{d-1} \tau^k C = 0.$$

Every continuation $\int_{\tau^k C} h$ equals the same single-valued function F . Averaging therefore gives

$$F = \int_{\frac{1}{d} \sum_k \tau^k C} h = 0.$$

Thus the kernel is zero. $\square$

If $F = \sum_\ell b_\ell(\mathbf{a}) v_\ell$ in a complex basis of its finite coefficient span in $\mathcal{V}_C$, Lemma 6.2 says that the $D_t^q v_\ell$ remain linearly independent. Consequently

$$(6.3) \quad \text{cont}_R(D_t^q F) = \text{cont}_R(F),$$

and (6.2) yields

$$(6.4) \quad \text{cont}_R(M_\mu) = \text{cont}_R(J_\mu).$$

Cayley–Hamilton recurrence. Work in the free rank- d $R[t]$ -algebra

$$\mathcal{A} = R[t, x]/(f(x) - t)$$

with basis $1, x, \dots, x^{d-1}$. Let T_g be multiplication by g_a . For arbitrary $\deg g$, reduction modulo the monic relation $f(x) - t$ gives a d -by- d matrix over $R[t]$. Write

$$\chi_g(Z) = Z^d + c_1(t; \mathbf{a})Z^{d-1} + \dots + c_d(t; \mathbf{a}).$$

Cayley–Hamilton, followed by the $R[t]$ -linear zero-cycle functional, gives for $\mu \geq d$

$$(6.5) \quad J_\mu + \sum_{i=1}^d c_i J_{\mu-i} = 0.$$

At $\mu = d$, the last term is zero because $J_0 = \int_C 1 = 0$. Multiplication by $H \in R[t]$ cannot enlarge parameter content: its R -coefficients multiply existing generators, while $t^k \int_C h = \int_C f^k h$ remains in $\mathcal{V}_C$. Hence

$$(6.6) \quad \text{cont}_R(HF) \subseteq \text{cont}_R(F).$$

Induction in (6.5) puts every $\text{cont}_R(J_\mu)$ in the ideal generated by the first $d - 1$ such contents. Equation (6.4) transfers the statement unchanged to the Melnikov sequence.

Theorem C (ordinary coefficient Bautin bound).

$$(6.7) \quad \boxed{\mathcal{B}_\infty = \mathcal{B}_{d-1}.}$$

Thus the ordinary coefficient Bautin index is at most $d - 1$ for every degree- d phase, every zero-cycle and every polynomial perturbation. In the degree box of [17],

$$(6.8) \quad \boxed{b_{\text{coeff}}(m) \leq m - 1.}$$

For a nonmonic phase, divide f, t , and g by the nonzero leading coefficient of f . This is a coefficient-ring automorphism together with multiplication of Melnikov coefficients by units, so it preserves the stabilization index and extends (6.8) to the full degree box.

Sharpness and convention boundary. Let $Z_0(f; \mathbf{Z}) = \{(n_1, \dots, n_d) \in \mathbf{Z}^d : \sum n_j = 0\}$ be the integral zero-cycle lattice, and put $Z_0(f; \mathbf{C}) = Z_0(f; \mathbf{Z}) \otimes_{\mathbf{Z}} \mathbf{C}$. For $f = x^d$, $x_j = \omega^j T$ and weights in the complexified module $n_j = \omega^j$, the Atlas DFT support is the singleton $\{d - 1\}$. With $g = ax$, (6.2) vanishes through order $d - 2$ and is nonzero at order $d - 1$. Hence (6.7) is sharp on the complexified cycle module $Z_0(f; \mathbf{C})$; this does not establish sharpness on the integral cycle lattice $Z_0(f; \mathbf{Z})$.

For a witness in the integral lattice, let p be the least prime divisor of d , put $q = d/p$, and take $n_j = p - 1$ for $j \equiv 0 \pmod{p}$ and $n_j = -1$ otherwise. Its Fourier support is $\{q, 2q, \dots, (p - 1)q\}$; with $g = ax$, the first nonzero order is $q = d/p$. This proves an integer-cycle lower bound but not integer-cycle sharpness of $d - 1$.

Proposition 6.1 identifies the parameter-content ideal with the ordinary local Taylor-coefficient Bautin ideal used in [18]. We denote the resulting degree-box index by $b_{\text{coeff}}(m)$. If the terse ideal chain in [17] is intended in this ordinary coefficient sense, then $b(m) = b_{\text{coeff}}(m)$ and Theorem C answers Problem 8.4. The source does not settle that ambient-ring interpretation;

alternative analytic-function-ring, pre-contraction radical, and singular-base-point conventions remain outside the claim.

EXACT COMPUTATIONAL EVIDENCE

The computations are deterministic falsification and replay evidence. They do not prove the all-degree statements.

Atlas-independent support/rank reconstruction. A separate exact checker imports no Atlas source. It constructs the normalized inverse by its coefficient recursion and compares observed non-trace residue channels with the exact rank of a standalone twisted-quotient Krylov matrix. Twenty-eight generic, composite, power, trace-only and Dickson cases pass through inverse-series depth 48. The checker first sums all monomial contributions at a Laurent exponent and examines only the exponent window complete at the declared depth. This prevents false channels from uncanceled partial expansions. The receipt witnesses nonzero coefficients in every observed channel; it does not certify vanishing beyond the finite window.

Dickson battery. One hundred seventy-one exact nondegenerate cases cover degrees three through nine at $a = 1, 2, -1/2$, both generators of every pair, nonconstant $\mathbf{Q}[P]$ -linear combinations and cross-pair unions. The predicted module support, Atlas inverse-series support and twisted-Krylov rank agree in every case. An $a = 0$ control loses the paired channel exactly as expected, confirming the excluded power degeneration.

Exceptional battery. Twenty-three exact cases cover all nine fused generators, within-module combinations, nonconstant $\mathbf{Q}[P]$ -coefficients, the two composition channels and cross-block unions. A separate Lagrange-inversion program checks all nine module/channel combinations through $\mathbf{Q}[P]$ -coefficient degree six using exact nonzero minors.

Weighted operator reconstruction. Two implementations agree coefficient for coefficient on seven cases of orders zero through four. The standalone implementation imports no Atlas code. It verifies (5.2) directly, and its verifier rejects a semantically modified operator even after the outer JSON digest is recomputed.

Representative operators, with coefficients ordered from low to high derivative, are

(P, A)	q	$(c_0, \dots, c_q)$
$(x^7, 1)$	1	$(1, 7s)$
$((x^4 + x)^3, 4x^3 + 1)$	1	$(1, 3s)$
$(x^5 - 5x^3 + 5x, 1)$	2	$(-100s^2 - 1, 25s, 25s^2).$

This is same-programme independent implementation, not independent human or institutional reproduction.

Bautin-bound checks. `check_general_bautin_bound.py` constructs fully generic quotient multiplication matrices with perturbation degree strictly above phase degree: $(d, n) = (2, 4)$ and $(3, 5)$. Exact arithmetic verifies Cayley–Hamilton and the zero-cycle infinity-monodromy average through degree twelve.

`screen_power_bautin_uniform.py` checks rectangular power-phase cases $(d, n) = (2, 4), (3, 5), (4, 6)$, representative coefficient ideals through order eight, and exact Fourier support and first-hit order for the lower witnesses. The checks deliberately exercise perturbations beyond the former degree boundary; neither finite program replaces the all-degree argument.

`check_bautin_content_bridge.py` uses the non-power phase $f = x^2 + x$, the cycle $x_1 - x_2$, and a generic quartic perturbation. Through order three it compares direct implicit-root coefficients

with (6.2), verifies equality of power-moment and differentiated Melnikov content, and recovers the same ideal from finite Taylor jets. All orders exercise dependent coefficient-function spans.

Package negative controls. The package-mutation verifier checks five distinct failure classes. It requires rejection of a modified manuscript byte stream, an undeclared root file, an unknown manifest field, a receipt whose embedded digest no longer matches, and a one-byte mutation of the frozen Atlas archive. Each mutation must fail with its expected diagnostic class; an unrelated exception is not a pass. These controls test the package gate, not the mathematical proofs.

DISCUSSION

The Dickson result turns pairwise Laurent symmetry into a uniform all-amplitude classification. Its parity effect is easy to miss if one checks only even degrees: odd phases have no midpoint and therefore cannot have odd rank.

The exceptional Ritt result shows a different fusion mechanism. The three correction-polynomial modules are not composition summands. Their rigidity comes from the irreducible standard representation of the quartic monodromy, while the outer cube imposes a modulo-three grading. The result is a complete support partition rather than a list of isolated examples.

The coefficient-Bautin theorem is the broadest of the three claims. Its content is not Noetherianity alone: Cayley–Hamilton gives a finite recurrence, and infinity monodromy shows that the order-dependent differentiation in the Melnikov formula preserves the ordinary parameter ideal exactly. This produces an explicit linear bound for the full degree box, not only a description of its centre set.

The weighted order theorem is conceptually useful even though its reduction algorithm is not new. Creative telescoping discovers the first dependence; the reconstructed support quantity predicts its order from inverse support or root values. Chen et al. [16] explain non-minimality of individual-contour telescopers through residues in the rational setting. Their result is a close discrete analogue of the residue obstruction below, not a prior statement of our complete-system support identity. That prediction is the narrow bridge proposed here.

Why raw channel count is not individual-period order. A stronger formula considered during development would make the minimal order of an individual period equal the number of residue channels left active by its cycle DFT. That statement is false.

For $P = D_3(x, 1) = x^3 - 3x$, $A = 1$, a complex zero-cycle can isolate one of the two residues of $G = x$. The isolated function nevertheless satisfies the modified Bessel equation

$$f'' + \frac{1}{s}f' - \left(4 + \frac{1}{9s^2}\right)f = 0,$$

which has no first-order factor over $\mathbf{C}(s)$. A rational logarithmic derivative would have residue $\pm 1/3$ at zero, residue 1 at every other finite pole, and total finite residue $-1/2$ from its expansion at infinity—an impossibility. Differentiation preserves the residue label but does not preserve the one-dimensional span of a channel function.

This adverse example is why no unrestricted equality with $|R|$ is claimed. It is consistent with the repaired bound (5.4): one touched channel lies inside a two-dimensional Dickson block, so the upper bound is two, and the explicit irreducible Bessel equation makes the actual order two.

LIMITATIONS AND OPEN OBLIGATIONS

1. The load-bearing support theorem is reproduced self-containedly, with a corrected Wronskian expansion, but the reconstruction followed the Atlas proof architecture and was performed within the same research programme. It has deterministic internal replay, not independent reconstruction or specialist proof review.
2. The novelty search is bounded. Exact-phrasal searches and the inspected primary corpus cannot establish priority.
3. No independent human specialist has reviewed the complete proofs.
4. The programs use exact rational inputs; algebraic or floating inputs are outside the retained certificate contract.
5. The standard-ray channel contact is proved by the formal-moment and Fourier–Gamma realization. Arbitrary contour systems and contour degeneracies remain outside the result.
6. The original exact active-channel formula was refuted. Only the corrected block bounds (5.4) are retained; a general Dickson block-mass equality is not claimed without a separate differential-module derivation.
7. Proposition 6.1 identifies the parameter-content ideal with the ordinary local Taylor-coefficient Bautin ideal. Alternative localized, radical, or analytic-parameter conventions remain outside the theorem.

These obligations prevent any claim of peer review, specialist validation or absolute priority. They do not invalidate a clearly labelled unrefereed candidate release after author approval and package review.

REPRODUCIBILITY AND DATA AVAILABILITY

The candidate research package contains:

- immutable source-provenance records and hashes;
- the reconstructed support theorem, the general block theorem and four full result dossiers;
- the bounded novelty audit;
- the binomial Laurent supplement and the retained adverse counterexample;
- exact falsification programs and JSON receipts, including an Atlas-independent support/rank reconstruction;
- the standalone weighted-order implementation and verifier;
- public theorem-specific adversarial review notes; and
- a manifest and offline integrity checker.

Research inputs are polynomial expressions and public scholarly sources. No personal-data analysis is performed. The immutable Atlas archive and frozen scholarly-source responses are referenced by path and SHA-256 in SOURCE_PROVENANCE.json but are not redistributed by this manuscript.

ETHICS STATEMENT

This is mathematical and computational research using public sources and synthetic polynomial inputs. No human participants, animals, clinical data, or personal data are involved. Ethical approval was not applicable.

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BINOMIAL LAURENT MOMENTS

This appendix records a useful corollary. Pakovich, Pech and Zvonkin [15] name and exemplify this exact Laurent polynomial moment-problem family. The corollary is not counted as one of the three residual novelty claims because its load-bearing classification is already in Pakovich [11].

Let

$$L(z) = z^{-m} + cz^n, \quad h = \gcd(m, n), \quad N = m + n.$$

For Laurent one-forms define

$$\mathcal{A}(L) = \{A(z) dz : \text{Res}_0 L^k A dz = 0 \text{ for every } k \geq 1\}.$$

Proposition A.1. The complete solution module is

$$\mathcal{A}(L) = \bigoplus_{\ell \not\equiv -1 \pmod{h}} \mathbf{C}[z, z^{-1}]_\ell dz \oplus d\mathbf{C}[L],$$

and

$$\text{rank}_{\mathbf{C}[L]} \mathcal{A}(L) = \frac{(h-1)N}{h} + 1.$$

In particular, if $\gcd(m, n) = 1$, then $\mathcal{A}(L) = d\mathbf{C}[L]$.

Proof. Rotation by an h -th root of unity fixes L and multiplies $z^\ell dz$ by $\omega^{\ell+1}$. Every non-residual grade therefore has all moments zero.

For the residual grade put $w = z^h$. The phase becomes

$$\tilde{L}(w) = w^{-m/h} + cw^{n/h},$$

with coprime exponents. Its critical points satisfy

$$w^{N/h} = \frac{m}{cn},$$

after cancellation of the common factor. They are simple, and their critical values are distinct because exponentiation by $-m/h$ is injective on the (N/h) -th-root orbit. Thus the connected monodromy group is transitive and generated by transpositions, hence is the natural symmetric group.

The phase is proper because it has poles at both zero and infinity. Its rational cover is connected, the finite critical points above are simple with distinct values, and their transpositions generate a transitive group; hence the group is $S_{N/h}$ and is doubly transitive. Pakovich’s Laurent-moment classification [11, Theorem 6.5] therefore applies with every hypothesis verified and says that the primitive residual-grade solutions are exactly $d\mathbf{C}[\tilde{L}]$. Pullback gives $d\mathbf{C}[L]$.

Finally, $\mathbf{C}[z, z^{-1}] dz$ is free of rank N over $\mathbf{C}[L]$, with a consecutive exponent window from $-m$ through $n - 1$. Each deck grade has rank N/h . The $h - 1$ automatic grades contribute $(h - 1)N/h$, and the exact residual summand contributes one. $\square$

No finite-test bound is stated here: its degree convention and exact specialization of Pakovich's orbit bound would require separate verification.